

Lab: Neural Correlates of the Self-Model

Learning Goal: Analyze the neural architecture of the self using the Active Inference framework.

Part 1: Mapping the Self-Model Network

Exercise: Map each component of the self-model to its neural substrate and the type of self-representation it implements:

Self-Model Component	Primary Brain Region	Type of Self-Representation	What Happens if Damaged
"What am I thinking/feeling?"	medial PFC	Reflective self	Difficulty understanding own mental states
"Who am I over time?"	Posterior cingulate	Narrative self (autobiography)	Loss of autobiographical memory
"How do others see me?"	TPJ / angular gyrus	Social self (perspective-taking)	Impaired theory of mind
"Where have I been?"	Hippocampus	Temporal self (episodic memory)	Amnesia, no temporal identity
"Where am I in space?"	Precuneus	Spatial self (first-person perspective)	Disorientation, OBE vulnerability
"What does my body feel like?"	Anterior insula	Interoceptive self (sentient self)	Alexithymia, emotional flatness

Now answer: Which components must be intact for the "minimal self" (basic pre-reflective awareness of being a subject)? Which are needed only for the "narrative self" (autobiographical identity)?

Write your response here

Part 2: Interoceptive Inference Thought Experiment

Learning Goal: Understand emotions as interoceptive prediction errors.

Exercise: Work through the following scenario step by step using interoceptive inference:

Scenario: You're about to give a public presentation. Here is a moment-by-moment Active Inference analysis:

Time	Body State	Brain's Prediction	Prediction Error	Emotion Felt	Action
T-5 min	Slight heart rate increase	"Heart should be at resting rate"	Moderate — HR above predicted	Mild anxiety	Internal — "It's just preparation" (model update)
T-2 min	Heart racing, dry mouth, sweaty palms	"Heart should be at resting rate"	LARGE — multiple interoceptive deviations	Strong anxiety	?
T-0	You step on stage, crowd smiles	?	?	?	?
T+5 min	You're in flow, voice steady	?	?	?	?
T+30 min	Presentation finished, applause	?	?	?	?

Complete the table for T-2 through T+30. For each row, specify: (a) the brain's prediction, (b) the prediction error, (c) the emotion felt, and (d) the action taken (belief update or physical action).

Write your response here

Part 3: Clinical Self-Model Analysis

Learning Goal: Apply Active Inference to clinical disorders of selfhood.

Exercise: For each clinical case, identify: (a) which component of the self-model is disrupted, (b) the Active Inference mechanism, and (c) a prediction for treatment.

Case 1 — Depersonalization: A 28-year-old reports "I look in the mirror and it doesn't look like me. I feel like I'm watching my life from outside my body. I can't feel my emotions properly."

- Which self-model component is disrupted?
- What is the Active Inference mechanism? (Hint: interoceptive precision)
- What treatment approach does this mechanism suggest?

Write your response here

Case 2 — Cotard's Syndrome: A 65-year-old stroke patient insists "I am dead. My organs have stopped working. I don't exist."

- Which self-model component is disrupted?
- What is the Active Inference mechanism? (Hint: catastrophic interoceptive inference failure)
- How does this relate to the anterior insula?

Write your response here

Case 3 — Phantom Limb: A veteran who lost his left arm reports: "I can still feel my arm. Sometimes it burns, sometimes it clenches into a fist. I know it's not there but I can FEEL it."

- Which self-model component is disrupted?
- What is the Active Inference mechanism? (Hint: prophylactic proprioceptive model)
- Why does mirror therapy work? (Explain using prediction error and model updating)

Write your response here

Part 4: The Rubber Hand Illusion

Learning Goal: Understand body ownership as Bayesian inference.

Exercise: Describe the rubber hand illusion step by step using Active Inference:

1. **Before stimulation:** The brain's body model includes the real hand. How confident is the model? (precision = high)
2. **During synchronous stroking:** Visual observation (seeing rubber hand being stroked) and tactile sensation (feeling your real hand being stroked) become correlated. What prediction error arises? How does the brain resolve the conflict?
3. **During asynchronous stroking (control condition):** Why does the illusion NOT occur? What is different about the prediction errors?
4. **After illusion onset:** Participants report a temperature drop in the real hand. What does this reveal about the brain's proprioceptive precision weighting? (It reduces precision for the real hand because the model has "reassigned" ownership)
5. **Extension:** Design a variation of the rubber hand illusion that would test a specific Active Inference prediction about body ownership. What would you change, and what would the result tell you?

Write your response here

Part 5: Meditation and the Self-Model

Learning Goal: Analyze contemplative practices through the lens of self-model modulation.

Exercise: Experienced meditators report: "The sense of being a fixed, separate self dissolves during deep meditation. The boundary between self and world becomes less rigid."

1. What would this look like in Active Inference terms? Which precision parameter(s) are being modulated?
2. Neuroimaging studies show reduced DMN activity during meditation. How does this connect to the Active Inference interpretation?
3. If the self is a prediction that can be temporarily suspended, what does this imply about the nature of selfhood?
4. Is meditation "delusional" (the self-model becomes less accurate) or "clarifying" (the self-model was always an approximation, and meditation reveals this)? Argue both sides.

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Network mapping	DMN regions and self-model components
2	Process tracing	Interoceptive inference moment by moment
3	Clinical analysis	Self-model disruptions (depersonalization, Cotard's, phantom limb)
4	Experimental reasoning	Rubber hand illusion as body ownership inference
5	Philosophical analysis	Meditation and self-model modulation